

12 M001 Week 3 — English Worksheet (Beginner)

Topic: Variables · **Course:** Pyramid Problems · **Level:** Beginner · **Time:** about 30 minutes

1 · Predict 🧠

Read the steps. Before you imagine the agent doing it, circle or write your answer.

```
set size to 3
repeat size times:
  place block down
  move forward
```

How many blocks does the agent place? Circle one: 1 · 3 · 6

Where does the number 3 come from?


```
set size to 6
repeat size times:
  place block down
  move forward
```

Same code, new number. How many blocks now? Circle one: 3 · 6 · 12

2 · Find the Difference 🐛

Each pair shows clean steps first, then a broken version of the same idea. Circle what's different and write one short sentence about the bug.

Pair A — The agent should place a row of **size** blocks, where size is 5.

```
# clean
set size to 5
repeat size times:
  place block down
  move forward
```

```
# buggy
set size to 5
repeat 3 times:
  place block down
  move forward
```

What is wrong? Which number does the loop really use?

Pair B — The agent should build a **tower** that is **height** blocks tall.

```
# clean
set height to 4
repeat height times:
  place block down
  move up by 1
```

```
# buggy
set height to 4
repeat height times:
  place block down
  move forward
```

What is wrong?

3 · Fill the Gap

The agent should build a row of 4 blocks using a variable. One line is missing. Fill it in using the word bank.

```
_____  
repeat size times:  
  place block down  
  move forward
```

Word bank: set size to 4 · move forward · repeat 4 times

Write the missing line:

4 · Tell Me What You Built 📷

Now switch to your homework world. Use a `size` variable to build a **size-3 wall**. Then change the variable to **6** and run it again. When you finish, come back here.

Send a photo or video of both walls, then explain what you did. Use these sentence starters — write 2 or 3 sentences.

I made a variable called ...

When I changed it to 6, the wall ...

A variable is useful because ...

5 · Record Your Walkthrough 🎥

Take a video on your phone (or a parent's phone) and show your walls. Talk like you are teaching a friend. Try to use these words: **variable**, **set**, **loop**, **repeat**.

1. Show the small wall and the big wall.
2. Read your variable line out loud.
3. Change the number one more time and show what happens.

Write what you will say in your video. You can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.